

Class 12: Population Scale Analysis Homework

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Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
dataset <- read.table("https://bioboot.github.io/bgg213_W19/class-material/rs8067378_ENSG000001000000.txt")
table(dataset$geno)
```

```
A/A A/G G/G
108 233 121
```

```
nrow(dataset)
```

```
[1] 462
```

The above shows the sample size for each genotype, the total dataset consisting of 462 datapoints.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

AGonly <- dataset %>%
  arrange(dataset$geno) %>%
  filter(geno == "A/G")

AAonly <- dataset %>%
  arrange(dataset$geno) %>%
  filter(geno == "A/A")

GGonly <- dataset %>%
  arrange(dataset$geno) %>%
  filter(geno == "G/G")

summary(AGonly)

```

sample	geno	exp
Length :233	Length :233	Min. : 7.075
N.unique :233	N.unique : 1	1st Qu.:20.626
N.blank : 0	N.blank : 0	Median :25.065
Min.nchar: 7	Min.nchar: 3	Mean :25.397
Max.nchar: 7	Max.nchar: 3	3rd Qu.:30.552
		Max. :48.034

```
summary(AAonly)
```

sample	geno	exp
Length :108	Length :108	Min. :11.40
N.unique :108	N.unique : 1	1st Qu.:27.02
N.blank : 0	N.blank : 0	Median :31.25
Min.nchar: 7	Min.nchar: 3	Mean :31.82
Max.nchar: 7	Max.nchar: 3	3rd Qu.:35.92
		Max. :51.52

```
summary(GGonly)
```

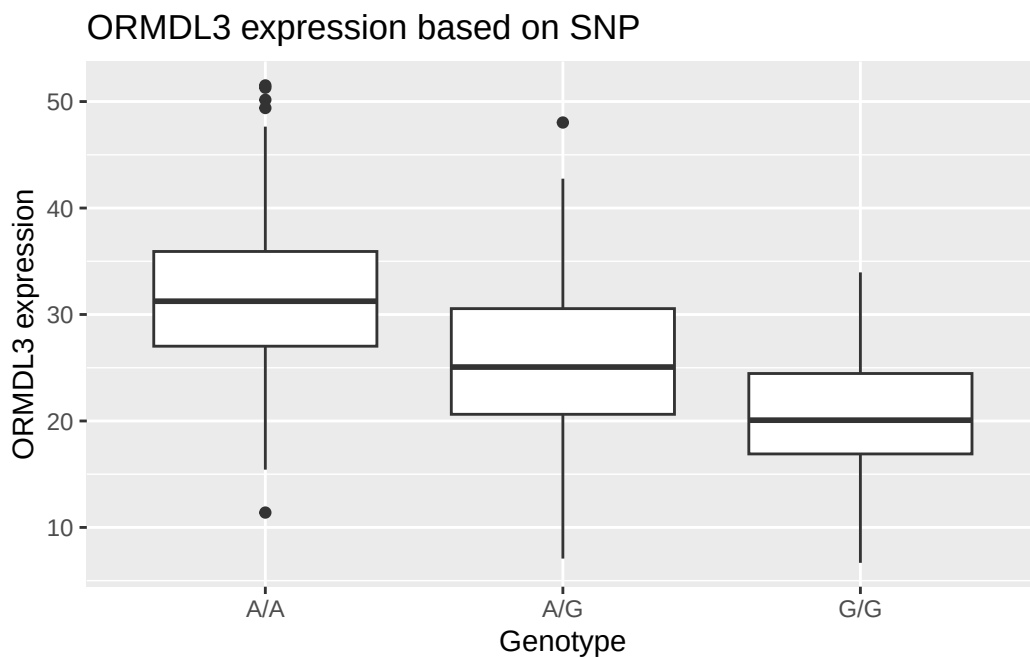
sample	geno	exp
Length :121	Length :121	Min. : 6.675
N.unique :121	N.unique : 1	1st Qu.:16.903
N.blank : 0	N.blank : 0	Median :20.074
Min.nchar: 7	Min.nchar: 3	Mean :20.594
Max.nchar: 7	Max.nchar: 3	3rd Qu.:24.457
		Max. :33.956

For A/G individuals the median expression for ORMDL3 is 25.065. For A/A, it is 31.25, and for G/G, it is 20.074.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)

ggplot(dataset, aes(x=geno, y=exp)) +
  geom_boxplot() +
  labs(y="ORMDL3 expression", x="Genotype", title="ORMDL3 expression based on SNP")
```



This SNP seems to affect the expression of ORMDL3, as G/G has much lower expression compared to A/A it seems, with A/G bridging the middle. Visually this looks like there is a decrease due to the G allele.